



Ministry of IT & Telecom
Government of Pakistan

FinMate



Pitch

We are developing Finmate, an AI-powered learning companion to help beginners and individual investors interested in stocks and cryptocurrencies understand trading strategies and market behavior through personalized lessons, real-time insights, and interactive simulations.

Introduction

BLOCK & STOCK

ORGANIZATION

Block and Stock

PROJECT TEAM

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THEMATIC AREA

EdTech + FinTech

OBJECTIVES

- Help beginners learn about **blockchain and stock investing** in one easy platform.
- Use **AI-powered lessons** to make difficult topics simple.
- Support **hands-on learning** with practice tools and live examples.
- Build **financial knowledge and confidence** for young learners.

The Problem

In Pakistan, most people have limited knowledge about stocks, cryptocurrency, and other investment options. This lack of awareness prevents many from making informed financial decisions and taking advantage of opportunities to grow their wealth. There is a need for a platform that simplifies these concepts and helps beginners learn investing in an easy and practical way.

In Pakistan, cryptocurrency ownership remains limited, with only 6.6% of the population owning digital assets as of 2023([Triple-A – Triple-A](#)). This low adoption rate indicates a significant gap in financial literacy and access to information regarding digital investments. Many individuals are either unaware of cryptocurrencies or find the concepts too complex to engage with confidently. This lack of understanding prevents them from exploring potential financial opportunities in the digital economy.

The Solution to the Problem

To address the lack of financial literacy and low cryptocurrency adoption in Pakistan, we propose a platform that teaches investing in a simple, step-by-step way. The platform will provide easy-to-understand lessons on stocks, cryptocurrencies, and other investment options. It will also offer practical tools such as simulations, quizzes, and real-time insights to help beginners practice what they learn. By simplifying complex concepts and guiding users through hands-on learning, the platform will help people gain confidence, make informed financial decisions, and gradually participate in the digital economy.

AI Wrapper Technologies

Feature	Description
Base Model Integration	Seamless integration with Groq and GPT-OSS-120B as the core LLM via a unified wrapper, providing consistent input-output handling.
Orchestration and Multiagent Support	Four specialized agents collaborate for a complete learning workflow: Tutor Agent (explains concepts), Practice Agent (quizzes, simulations, paper trading), Market Agent (real-time stock/crypto trends), and Ethics Agent (risks, scams, ethical investing).
Data Handling & Pre/Post Processing	Chats, user profiles, and activity are stored in a Supabase database with Argon2 encryption for passwords; ORM ensures protection against SQL injection.
Evaluations/Benchmarking Module	Tracks user learning progress and provides personalized pathways (predefined and custom) to optimize learning outcomes.
Scalability and Deployment Readiness	Optimized for on-premises deployment or cloud servers (AWS-like environments) for reliable, scalable performance.
Security & Compliance	OAuth2 authentication with Google and Microsoft; data stored privately and accessible only to the user; Kong API Gateway limits abuse and enhances security; prompt moderation guard prevents malicious input.
Extensibility	Segmented design allows seamless integration of new features, UI improvements, or additional agents without disrupting existing workflows.
Monitoring & Analytics	Comprehensive logging via Kong and custom logs tracks system performance, individual user activity, and errors, with dashboards visualized in Grafana.

Live Demo

The screenshot displays the FinMate AI Stock & Crypto Learning Companion interface. At the top, the header includes the FinMate logo and the subtitle "AI Stock & Crypto Learning Companion". Below this, a navigation bar contains three buttons: "AI Tutor", "Markets", and "Learn". The main content area is titled "AI Tutor" and features a chat interface with two user avatars: "Abdul Roman Urdu" (labeled 'A') and "John English Only" (labeled 'J'). The chat history shows a series of questions and answers from John. The first message from John is: "Hello! I'm John, your AI financial tutor. Ready to learn about stocks and crypto? Let's start with the basics!". Below this is a "Play Voice" button. The second question is: "What is the difference between stocks and crypto?". The response from the AI tutor is: "Stocks are shares that give you ownership in a company, while crypto are digital tokens that function as a currency or asset and don't represent company ownership.", followed by another "Play Voice" button. The third question is: "If I have getting into crypto and stocks what should I invest in crypto or stocks and why?". The response is: "Start with stocks for a more stable, long-term foundation, and add a small, affordable slice of crypto for higher-growth potential. Diversify and only invest money you can afford to lose.", with a "Play Voice" button. The fourth question is: "Can you specify exact stocks that I should invest in that have low risk and average to high profit margin?". The response lists several stocks: "Consider low-risk, high-margin stocks such as **Clorox (CLX), MSA Safety (MSA), MSCI (MSCI), Church & Dwight (CHD), American States Water (AWR), Linde plc (LIN), ADP (ADP), Sherwin-Williams (SHW), Snap-On (SNA) and Rollins (ROL)** – they have solid P/E ratios, analyst-rated upside and historically low volatility. These companies combine stable cash flows, strong brand power and consistent profitability, making them suitable for a conservative, profit-focused portfolio.", with a "Play Voice" button. At the bottom, there is a text input field with the placeholder "Ask me anything..." and two icons for voice and image input.

Impact

- Empowers users to understand stocks, crypto, and investing through personalized lessons.
- Provides real-time market insights and ethical guidance for informed decisions.
- Encourages responsible investing, reducing risks and promoting disciplined habits.
- Boosts personal wealth and foreign capital inflows, supporting Pakistan's economy.
- Promotes financial literacy, inclusion, and accessibility in English and Urdu.

Thank You!